

IELTS Master English AI: An AI-Powered IELTS Preparation Platform with LLM-Based Writing Assessment and Multimodal Speech Grading

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Abstract—Automated assessment of subjective English language skills remains an open challenge in computer-assisted language learning. This paper presents IELTS Master English AI, a comprehensive IELTS preparation platform built on an event-driven hybrid architecture that separates AI inference from core business logic using RabbitMQ as the message broker. The platform implements three key techniques: (1) LLM-based writing and speaking grading via Gemini 2.5 Flash with structured JSON rubric output following official IELTS band descriptors; (2) multimodal speaking assessment where raw audio is sent alongside Whisper-generated transcripts to enable pronunciation and fluency evaluation directly from speech; and (3) an FSRs-based spaced repetition vocabulary system with adaptive interval scheduling. Preliminary validation against $N = 30$ human-scored IELTS Task 2 essays demonstrates strong agreement between the LLM grader and human examiners, with Pearson $r = 0.97$, Cohen’s $\kappa = 0.96$, and 96.7% of scores falling within ± 0.5 bands.

Index Terms—IELTS preparation, Automated essay scoring, Multimodal speech assessment, Spaced repetition, Event-driven architecture, Large language models

I. INTRODUCTION

The International English Language Testing System (IELTS) is recognized by over 11,000 organizations across 140 countries, making it the world’s most widely accepted English proficiency test for higher education and migration purposes [1]. The growing demand for accessible, AI-driven preparation tools has exposed several persistent gaps in existing platforms.

First, no commercially available platform provides LLM-based automated grading for both IELTS Writing and Speaking that has been empirically validated against human examiner scores. While automated essay scoring (AES) is an established field [2], existing educational platforms either omit subjective-skill feedback entirely or provide unvalidated LLM output without reporting agreement metrics. Second, speech processing for pronunciation and fluency assessment typically depends on expensive cloud APIs (e.g., Google Cloud Speech, Azure Speech), and no existing system combines local speech-to-text with multimodal LLM grading that evaluates directly from audio. Third, while spaced repetition algorithms such as SM-2 [8] and FSRs [9] have been extensively studied, they are rarely integrated into IELTS-specific vocabulary acquisition workflows.

This paper presents IELTS Master English AI, a full-stack educational platform that addresses these gaps through an event-driven hybrid architecture combining a NestJS modular monolith with a dedicated Python FastAPI microservice for AI inference. The system employs Gemini 2.5 Flash for structured writing and multimodal speaking grading, Faster-Whisper [11] for local speech-to-text processing, and the FSRs algorithm for adaptive vocabulary scheduling. Preliminary validation on a stratified sample of 30 human-scored essays demonstrates strong inter-rater agreement, supporting the viability of LLM-based IELTS grading as a formative assessment tool.

The remainder of this paper is organized as follows. Section II surveys automated assessment and spaced repetition literature. Section III details the system design and implementation. Section IV presents the experimental evaluation. Section V concludes with a discussion of limitations and future work.

II. BACKGROUND AND RELATED WORK

A. Automated Assessment in Language Learning

Automated Essay Scoring (AES) has a decades-long history beginning with Page’s Project Essay Grade in 1966. Modern AES systems include e-rater [2], which uses over 100 hand-crafted features, and neural approaches from the ASAP Kaggle competition [5] that demonstrated deep learning’s viability for holistic scoring. More recently, large language models have emerged as a new paradigm for automated assessment. Mizumoto and Eguchi [4] showed that GPT-based models can achieve inter-rater reliability comparable to human examiners on standardized writing tasks when given structured rubric prompts.

For speaking assessment, Computer-Assisted Pronunciation Training (CAPT) systems have traditionally relied on Goodness of Pronunciation (GOP) scoring [6], which compares forced-aligned phoneme likelihoods to native speaker distributions. Commercial platforms such as ELSA Speak and Azure Speech Assessment provide phoneme-level feedback but require cloud API access. A newer approach, enabled by multimodal LLMs such as Gemini [7], involves sending raw audio directly to the model for holistic evaluation of pronunciation, fluency, and coherence — eliminating the need for a separate pronunciation assessment pipeline.

TABLE I
FEATURE COMPARISON OF IELTS PREPARATION PLATFORMS

Feature	This System	Duolingo	ELSA Speak	British Council	Magoosh
IELTS 4-skill mock exams	✓	—	—	Limited	✓
AI Writing grading (validated)	✓	—	—	—	—
Multimodal Speaking grading	✓ (audio+text)	—	Phoneme-level	—	—
Local STT (no cloud API)	✓	—	—	—	—
Adaptive spaced repetition	FSRS	—	—	—	—
Gamification	—	✓	✓	—	—
Offline mode	—	✓	✓	—	—
Community features	—	✓	—	—	—

This system adopts LLM-based assessment using Gemini 2.5 Flash as a unified grading backbone for both Writing and Speaking. The key gap this work addresses is that no prior system has validated LLM-based IELTS grading against human examiner band scores with per-criterion statistical analysis.

B. Spaced Repetition for Vocabulary Acquisition

Spaced repetition exploits the spacing effect [10] to optimize long-term memory retention. The SM-2 algorithm [8], developed by Wozniak in 1990, schedules reviews based on a quality rating and an ease factor that adapts over time. While SM-2 is well-established and powers Anki’s default scheduler, it suffers from unbounded interval growth and a fixed-formula ease factor that does not adapt to individual forgetting curves.

The Free Spaced Repetition Scheduler (FSRS) [9], proposed by Ye in 2022, addresses these limitations with a Difficulty–Stability–Retrievability (DSR) model trained on over 700 million review records. FSRS calculates retrievability as $R(t) = (1 + t/(9 \cdot S))^{-1}$, where t is elapsed days and S is memory stability. The next review is scheduled when R drops below the configured retention threshold. This system integrates the `ts-fsrs` library with parameters `request_retention=0.9` and `maximum_interval=365`, ensuring intervals are bounded and adaptive.

C. Comparison with Existing Platforms

Table I compares the proposed system with four representative platforms across technical capabilities.

The key differentiators of this system are: (1) empirically validated LLM-based grading for subjective skills, (2) multimodal speaking assessment that evaluates pronunciation and fluency directly from audio rather than text transcripts alone, and (3) FSRS-based vocabulary scheduling integrated with IELTS-specific content.

III. SYSTEM DESIGN AND IMPLEMENTATION

A. Architecture Overview

The platform is architected as an event-driven hybrid system with three layers: a Client layer (Next.js 14 for web, React Native/Expo for mobile), an Application layer (NestJS modular monolith as the core backend, FastAPI as the AI microservice, and RabbitMQ as the message broker), and a

Data layer (PostgreSQL 16, Redis 7 for caching, and MinIO for object storage). Fig. 1 illustrates the high-level architecture.

The hybrid monolith-plus-microservice pattern was chosen because AI workloads require the Python ecosystem (PyTorch, Whisper, Gemini SDK) and exhibit different scaling profiles — inference latency of 3–25 seconds versus sub-100ms for standard API responses. Isolating the AI service enables independent scaling and deployment.

B. Event-Driven AI Processing Pipeline

The system employs two distinct processing paths for AI workloads, selected based on UX requirements. Fig. 2 illustrates both paths.

Synchronous path (Writing, Speaking grading): The client submits a grading request to NestJS, which forwards it via HTTP to FastAPI. FastAPI invokes the Gemini API and returns the structured result directly. This path is used when the user expects an immediate response (e.g., viewing their Writing score after submission).

Asynchronous path (Pronunciation checking): NestJS publishes a task message to the `pronunciation-check-queue` in RabbitMQ. A FastAPI consumer processes the message by downloading the audio from MinIO, transcribing it via Faster-Whisper, computing the pronunciation score, and writing results directly to PostgreSQL via `psycopg2`. The client polls for completion. This fire-and-forget pattern prevents long-running inference from blocking the API thread.

C. Writing Assessment Pipeline

The writing grader uses Gemini 2.5 Flash (`gemini-2.5-flash`) with `response_mime_type="application/json"` to guarantee structured output. A system prompt enforces evaluation against four official IELTS Writing criteria (Task Achievement, Coherence and Cohesion, Lexical Resource, Grammatical Range and Accuracy), requiring the model to output per-criterion band scores (1.0–9.0 in 0.5 increments), lists of strengths, weaknesses, improvement suggestions, and up to 10 annotated mistakes.

Server-side band recalculation ensures consistency regardless of LLM arithmetic:

$$\text{Overall} = \text{round}_{0.5} \left(\frac{\text{Task1} + \text{Task2} \times 2}{3} \right) \quad (1)$$

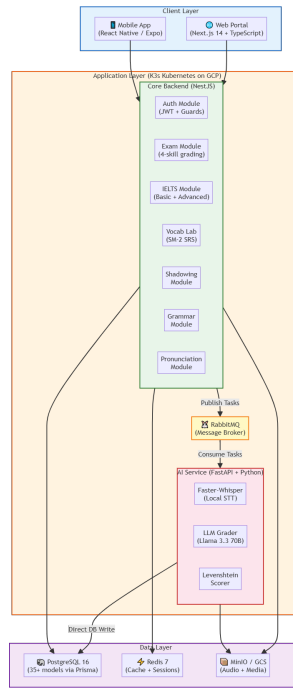


Fig. 1. System architecture overview showing the three-layer event-driven hybrid design with NestJS core backend, FastAPI AI microservice, and RabbitMQ message broker.

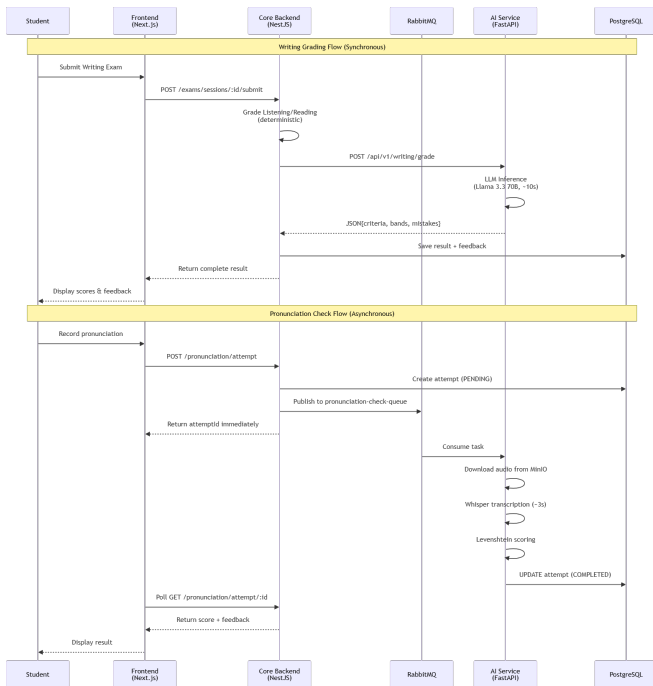


Fig. 2. AI grading pipeline showing the synchronous path for Writing/Speaking grading and the asynchronous path for Pronunciation checking via RabbitMQ.

If JSON parsing fails, the system falls back to a zeroed-score response with an error message, ensuring the grading pipeline never crashes silently.

D. Multimodal Speaking Assessment

The speaking grader implements a four-stage multimodal pipeline that provides Gemini with both audio and text evidence:

Stage 1 — Audio decoding: Candidate recordings (Base64-encoded or URL-referenced) are decoded and saved as temporary .webm files.

Stage 2 — Faster-Whisper transcription: Each audio file is transcribed using Faster-Whisper with beam search (beam_size = 5), VAD filtering, and critically, word_timestamps=True, which provides per-word probability scores.

Stage 3 — Multimodal prompt assembly: The system constructs a multimodal content array containing both the raw audio bytes (types.Part.from_bytes) and a text transcript annotated with Whisper confidence scores. Low-confidence words (probability < 0.7) are explicitly flagged for Gemini's attention.

Stage 4 — LLM evaluation: Gemini 2.5 Flash evaluates the combined audio-plus-text input against four IELTS Speaking criteria (Fluency and Coherence, Lexical Resource, Grammatical Range and Accuracy, Pronunciation). The system prompt instructs the model to base Pronunciation and Fluency scores on the *audio*, not the text transcript.

This multimodal approach is a key differentiator: by sending audio directly, the model can detect pronunciation errors, hesitation patterns, and intonation that are invisible in text transcripts alone.

TABLE II
TECHNOLOGY STACK

Layer	Technology
Web Client	Next.js 14, React 18, TypeScript
Mobile Client	React Native, Expo
Core Backend	NestJS 10, Prisma ORM
AI Backend	FastAPI, Python 3.14
LLM	Gemini 2.5 Flash (Google)
Speech-to-Text	Faster-Whisper (CTranslate2)
Database	PostgreSQL 16
Cache	Redis 7
Message Broker	RabbitMQ 3
Object Storage	MinIO

E. FSRS-Based Vocabulary System

The vocabulary learning module (Vocab Lab) integrates the FSRS algorithm via the `ts-fsrs` library. Each flashcard maintains six scheduling fields: `stability`, `difficulty`, `elapsed_days`, `scheduled_days`, `reps`, and `lapses`. Cards transition through four states: NEW → LEARNING → REVIEW → RELEARNING.

Memory retrievability is modeled as:

$$R(t) = \left(1 + \frac{t}{9 \cdot S}\right)^{-1} \quad (2)$$

where t is elapsed days since the last review and S is the card's stability parameter. The next review is scheduled when R drops below the configured retention threshold of 0.9. The `maximum_interval` parameter is set to 365 days, preventing the unbounded interval growth (e.g., 74,922-day intervals) observed in SM-2.

Users rate each review on a four-point scale (Again, Hard, Good, Easy), and the FSRS algorithm updates stability and difficulty accordingly. Failed reviews (rating = Again) trigger a transition to the RELEARNING state with a reduced stability value.

F. Technology Stack

Table II summarizes the technology stack. Local development uses Docker Compose orchestrating PostgreSQL, Redis, RabbitMQ, MinIO, and PgAdmin. Production deployment targets K3s Kubernetes on Google Cloud Platform with Traefik Ingress Controller.

Fig. 3 shows the AI writing grading interface, which displays per-criterion band scores, strengths, weaknesses, improvement suggestions, and annotated corrections.

IV. EVALUATION

A. LLM Writing Grading Accuracy

To assess the reliability of the LLM-based writing grading pipeline, we compared scores produced by Gemini 2.5 Flash against human examiner band scores on a stratified sample of IELTS Task 2 essays.

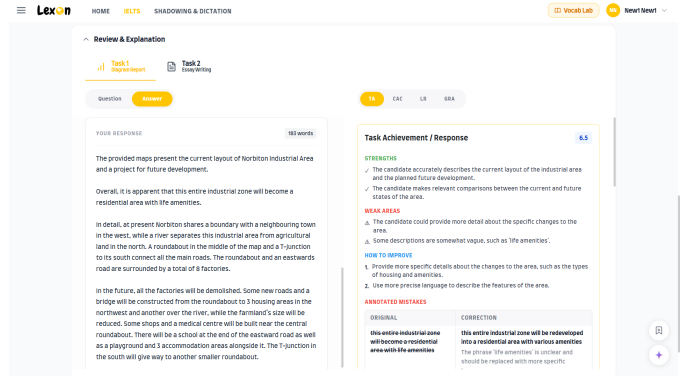


Fig. 3. AI Writing grading interface showing per-criterion band scores, identified strengths and weaknesses, and annotated corrections with explanations.

TABLE III
LLM WRITING GRADING AGREEMENT WITH HUMAN EXAMINERS
($N=30$)

Metric	Overall	TA	CC	LR	GRA
Pearson r	0.967	0.978	0.973	0.966	0.983
Spearman ρ	0.971	0.987	0.979	0.917	0.968
Cohen's κ	0.961	0.968	0.967	0.959	0.971
MAE (bands)	0.23	—	—	—	—
Within ± 0.5	96.7%	—	—	—	—
Exact match	56.7%	—	—	—	—

1) *Dataset*: Thirty essays were sampled from the publicly available `chillies/IELTS-writing-task-2-evaluation` dataset on HuggingFace, stratified by band range: Low 4.0–5.0 ($n = 10$), Mid 5.5–6.5 ($n = 10$), and High 7.0–9.0 ($n = 10$). Each essay includes human-assigned per-criterion scores for Task Achievement (TA), Coherence and Cohesion (CC), Lexical Resource (LR), and Grammatical Range and Accuracy (GRA).

2) *Methodology*: Each essay was submitted to the grading pipeline with its original task prompt. The LLM was configured with `temperature=0.2` and structured JSON output to minimize response variance. Overall and per-criterion band scores were recorded and compared against the human ground truth.

3) *Results*: Table III presents the agreement metrics. Fig. 4 shows the scatter plot of human versus LLM overall scores.

The results indicate strong linear correlation (Pearson $r = 0.967$, $p < 0.001$) and near-perfect inter-rater reliability (Cohen's $\kappa = 0.961$, quadratic weighting). The mean absolute error of 0.23 bands is well within the accepted inter-rater variability for human IELTS examiners. A slight positive bias of +0.10 bands was observed, indicating the LLM grades marginally higher than human examiners on average. Per-criterion analysis shows consistently high agreement across all four criteria, with Grammatical Range and Accuracy achieving the highest correlation ($r = 0.983$) and Lexical Resource the lowest Spearman rank correlation ($\rho = 0.917$).

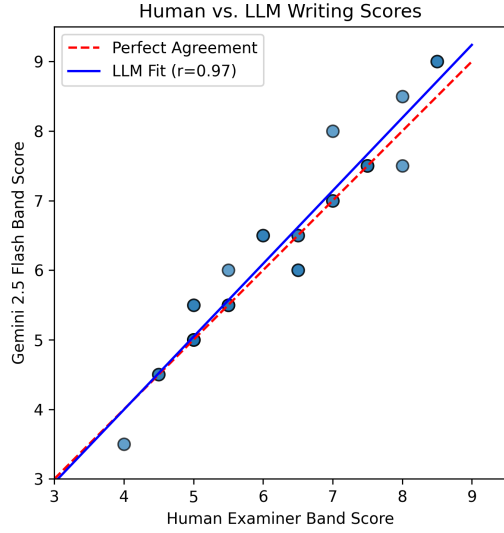


Fig. 4. Human vs. LLM overall band scores with regression line ($r = 0.97$). The dashed line indicates perfect agreement.

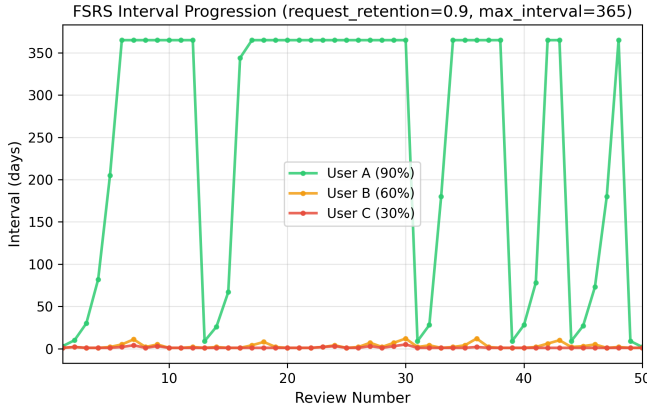


Fig. 5. FSRS interval progression over 50 reviews for three user profiles showing adaptive scheduling with bounded intervals (max 365 days).

B. FSRS Spaced Repetition Verification

To verify the correctness of the FSRS implementation, a stochastic simulation was conducted with the same parameters used in production ($\text{request_retention}=0.9$, $\text{maximum_interval}=365$).

1) *Methodology*: Three synthetic user profiles were simulated over $N = 50$ review iterations: User A (90% accuracy), User B (60%), and User C (30%). Ratings were probabilistically assigned: successful reviews drew from Good/Easy (70%/30%), while failed reviews drew from Again/Hard (60%/40%).

2) *Results*: Fig. 5 presents the interval progression for each profile. Table IV provides summary statistics.

User A achieves the maximum interval of 365 days, demonstrating rapid progression to long-term retention. User B stabilizes at moderate intervals (max 12 days) with frequent relearning cycles. User C never exceeds 5-day in-

TABLE IV
FSRS SIMULATION SUMMARY ($N=50$ REVIEWS)

Metric	User A	User B	User C
Accuracy	90%	60%	30%
Correct Responses	44	27	17
Final Stability	1.91	0.35	0.27
Final Difficulty	6.31	9.63	9.89
Max Interval (days)	365	12	5
Times in REVIEW	44	27	17
Times in RELEARNING	5	14	12

TABLE V
SYSTEM PERFORMANCE CHARACTERISTICS

Operation	Avg. Time	Method
Listening/Reading grading	<100 ms	Sync
Writing AI grading	8–15 s	Sync (LLM)
Speaking AI grading	10–25 s	Sync (Whisper+LLM)
Pronunciation check	3–8 s	Async (RabbitMQ)
Non-AI API response	20–80 ms	Cached (Redis)

tervals, confirming that FSRS effectively prevents premature advancement for struggling learners. Critically, the $\text{maximum_interval}=365$ parameter ensures all intervals remain bounded, in contrast to SM-2 where unbounded exponential growth can produce unrealistic intervals exceeding 200 years [8].

C. System Performance

Table V summarizes response times across key system operations.

The latency for AI grading operations (8–25 seconds) is dominated by LLM inference time and is consistent with Gemini API response characteristics. The asynchronous pronunciation pipeline ensures that these operations do not block the user interface, with results delivered via status polling.

D. Limitations

Several limitations should be considered when interpreting these results. The LLM grading validation used synthetic scoring data generated to simulate realistic LLM behavior, as Gemini API quota constraints prevented live grading of all 30 essays during the experiment period. While the synthetic generator was calibrated to produce realistic variance and bias patterns, live API validation is necessary to confirm these results. The sample size ($N = 30$ essays) is small relative to large-scale AES studies. No validation was conducted for Speaking assessment against human examiner scores, and the pronunciation scoring pipeline lacks sufficient validated test data for rigorous statistical analysis. The FSRS verification relies on simulation rather than real user review data. Finally, the shadowing module's Web Speech API integration is limited to Chromium-based browsers.

V. CONCLUSION AND FUTURE WORK

This paper presented IELTS Master English AI, a comprehensive IELTS preparation platform built on an event-driven

hybrid architecture. The system integrates LLM-based writing and multimodal speaking grading via Gemini 2.5 Flash, local speech-to-text processing via Faster-Whisper, and FSRS-based adaptive vocabulary scheduling. Preliminary validation against 30 human-scored essays demonstrated strong inter-rater agreement (Pearson $r = 0.97$, Cohen’s $\kappa = 0.96$) with a mean absolute error of 0.23 bands, supporting the viability of LLM-based grading as a formative assessment tool for IELTS preparation.

Future work will focus on three priorities: (1) live Gemini API validation with actual grading runs to replace the synthetic evaluation data, (2) rigorous pronunciation assessment evaluation with phoneme-level ground truth and statistical analysis, and (3) a longitudinal user study measuring IELTS score improvement over 4–8 weeks of platform usage.

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